

Timing Analysis

Timing analysis is the process of verifying that a circuit design meets its timing constraints, ensuring signals arrive at their destination within specified time limits. It involves checking for setup and hold time

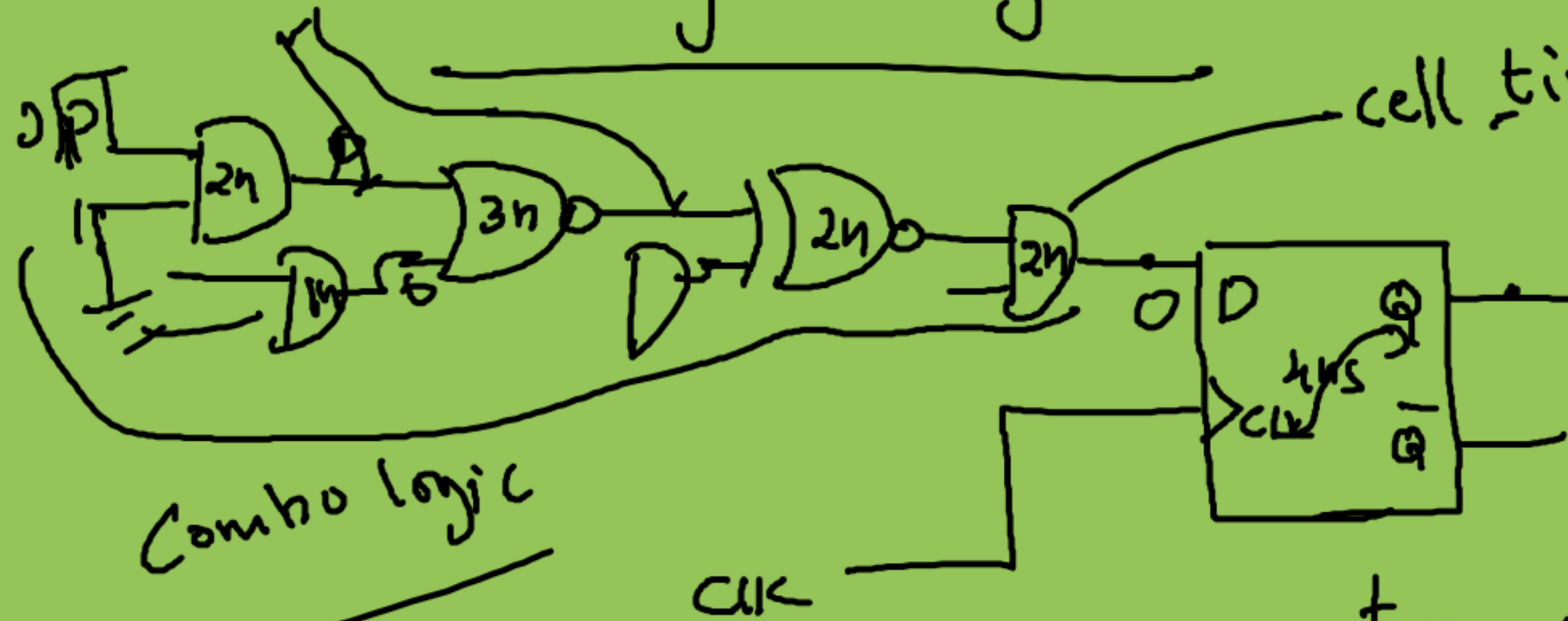
Purpose

- To ensure that **data signals** arrive at flip-flops, latches, or output pins **at the right time** relative to the clock.
- To confirm that there are **no setup or hold violations**, which can cause functional errors.

Types of Timing Analysis

Type	Description
Static Timing Analysis (STA)	Calculates the worst-case path delays in the design without using input test vectors. It checks all possible timing paths mathematically.
Dynamic (Simulation-Based) Timing Analysis	Uses test vectors to simulate real switching activity and measure actual delays during operation.

net timing arc Timing analysis



cell timing arc

Library
files
Lib.file

$$2n + 3n + 2n + 2n + 4ns$$

13ns

t_{clk-q} delay

Timing analysis

Divide circuit into path + add propagation delay

The tool will check that with in 13 ns the data is coming or not

if not verify timing violation.

.lib file

setup
Hold up

Liberty file - The parameter that are
- cell name, Input transition time, output load capacitance
delay time, Transition time, Capacitance (parasitic element)

net timing arc $\boxed{Z = RC}$ (parasitic element)

Static timing analysis

- * check every path in the design for timing violation
- * EDA Tools (Electronic design automation)
- * tempus, cadence Design system.

Dynamic Timing analysis

- Applies input vector & check for the correct o/p vector.
- * ModelSim, Mentor graphics

STA

* Doesn't verify functionality.

(STA doesn't check for logical correctness of the design)

* Faster

* Good coverage as check all path

DTA

* Verify functionality.

* slower

* less coverage, access depend on the test stimulus.

STA

- * less accurate
- * check synchronous part of Design
- * Can't detect timing error related to logical operation
- * Can report false error

DTA

- * More accurate
- * check both asynchronous & synchronous.
- * Input stimulus applied can detect timing error like race around condition
- * Doesn't report false error.

STA

DTA

Worst case, best
case analysis

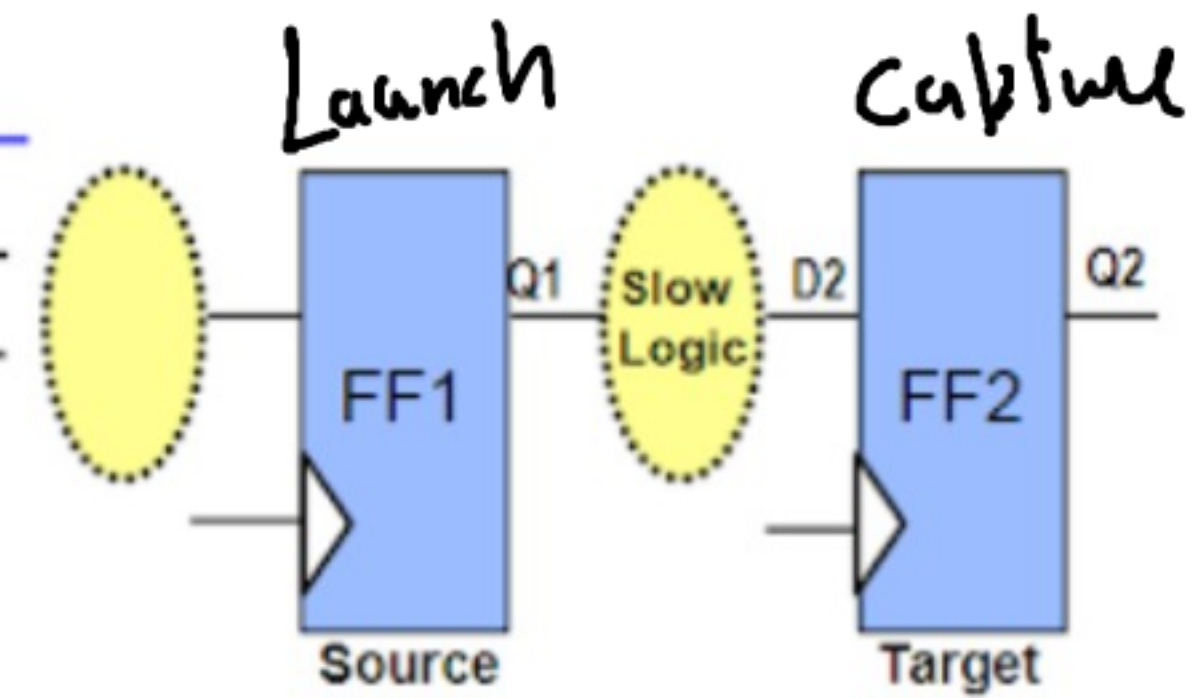
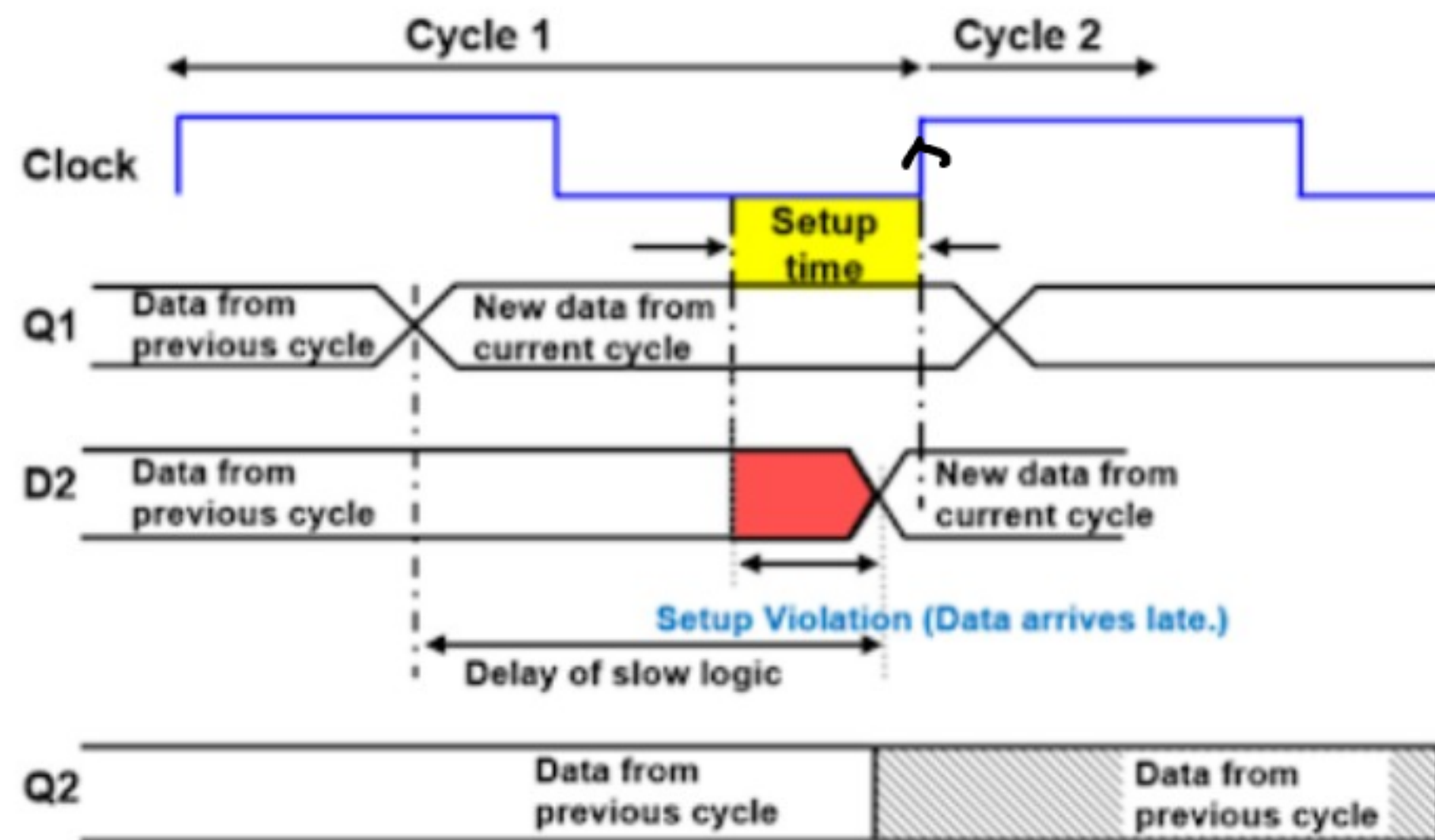
Complexity in finding
the input stimulus that
produce maximum delay.

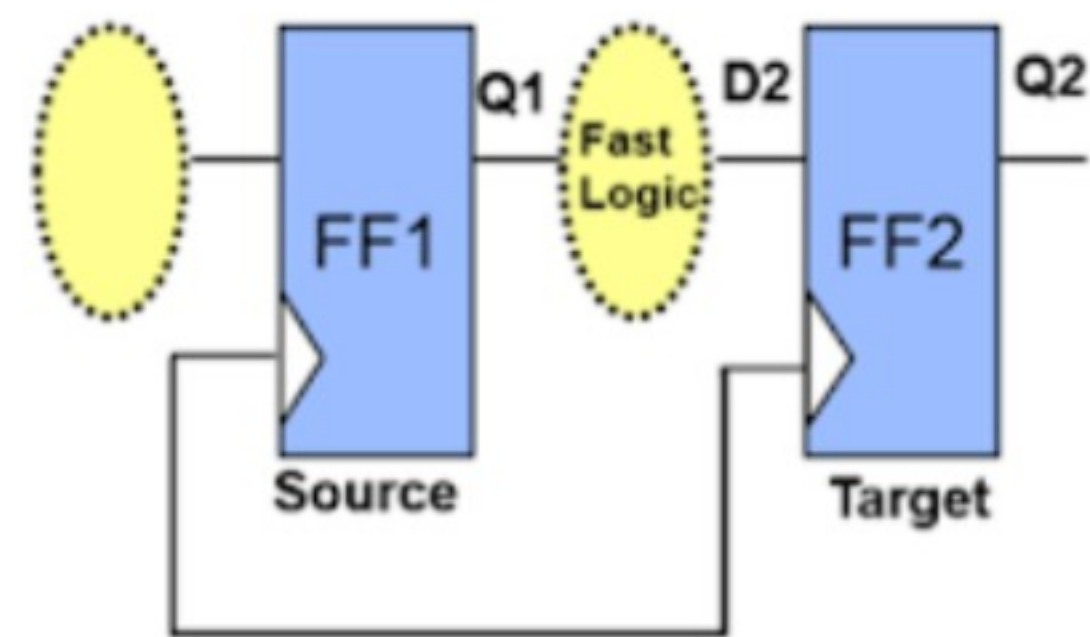
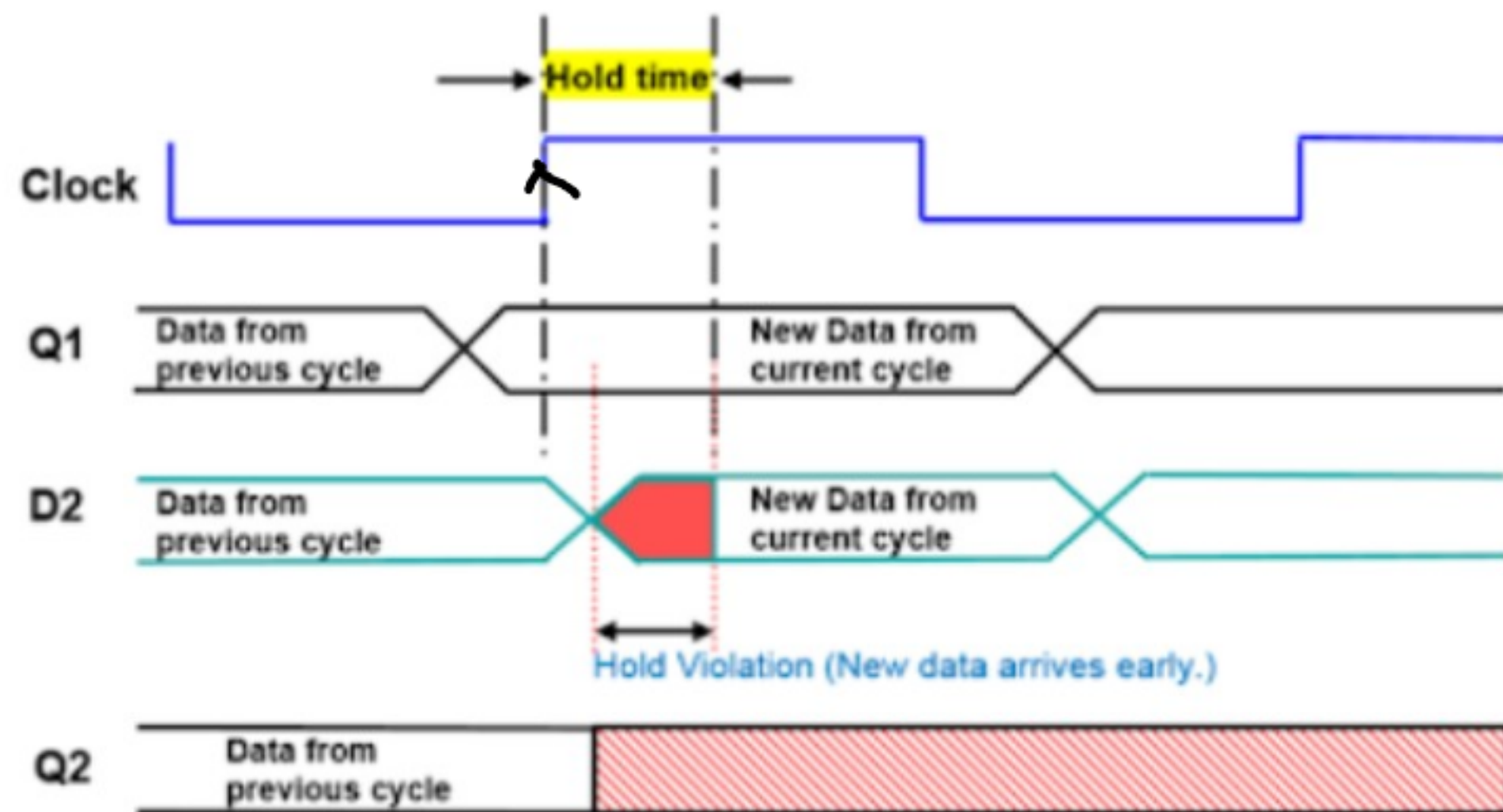
Clocked storage element

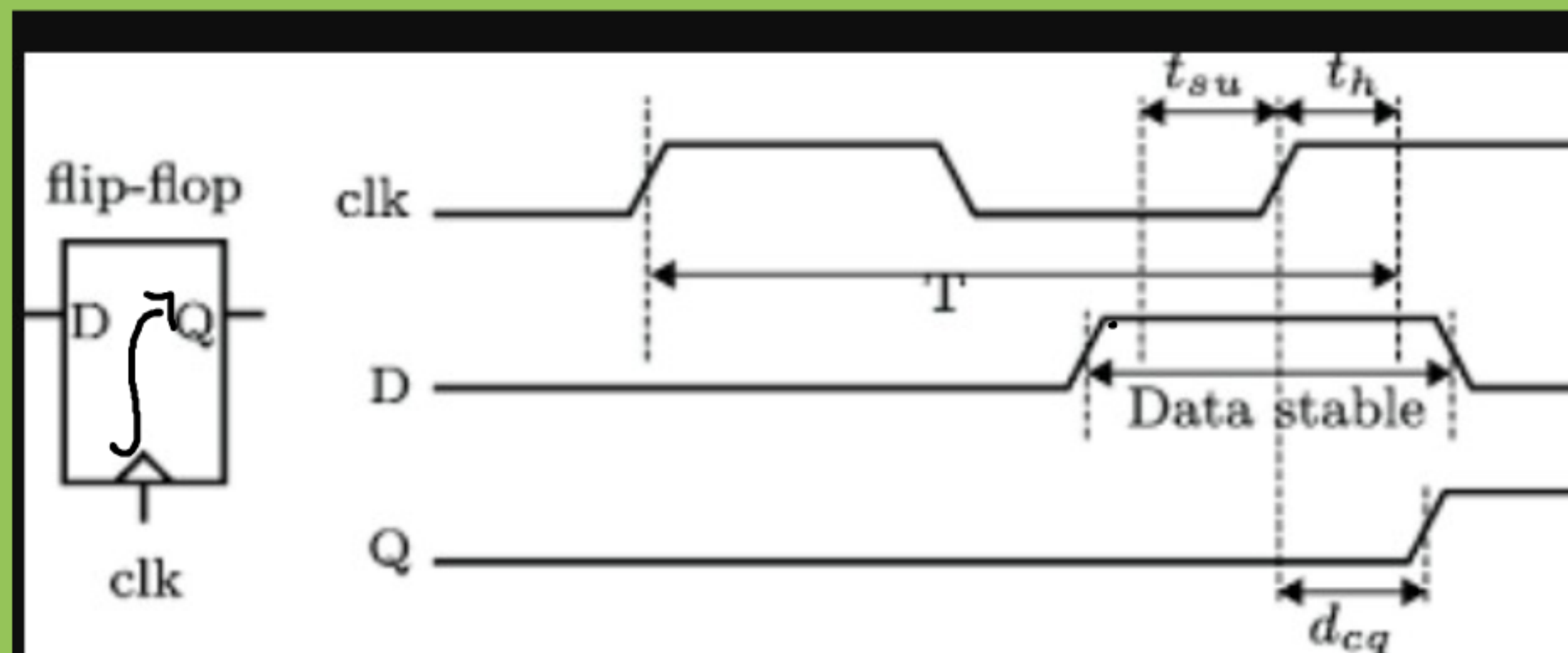
Advantage of Flip flops

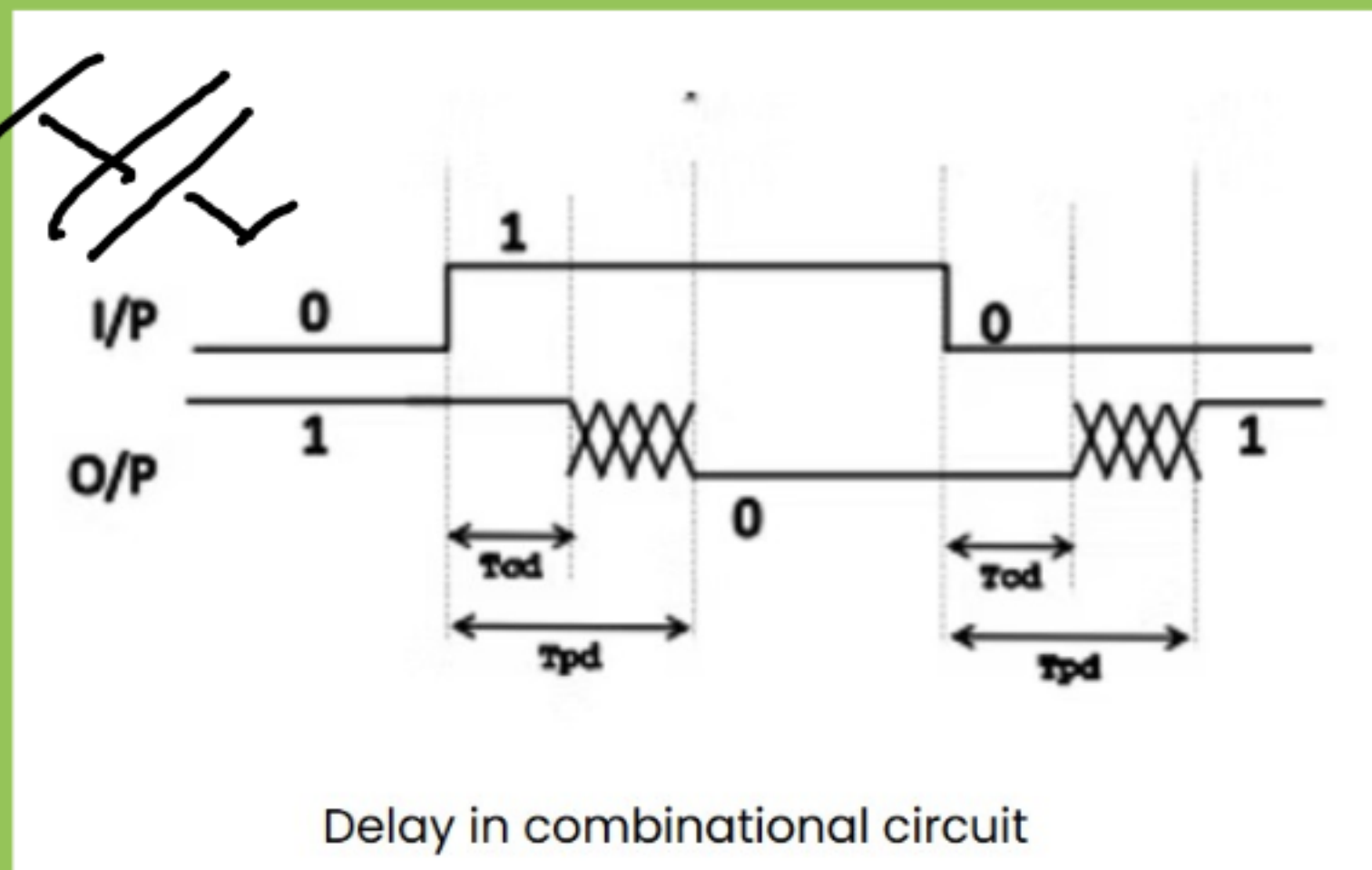
- * Synchronization : Precise timing control to digital circuits.
- * Stability : Better noise immunity & less susceptible to glitch.

*







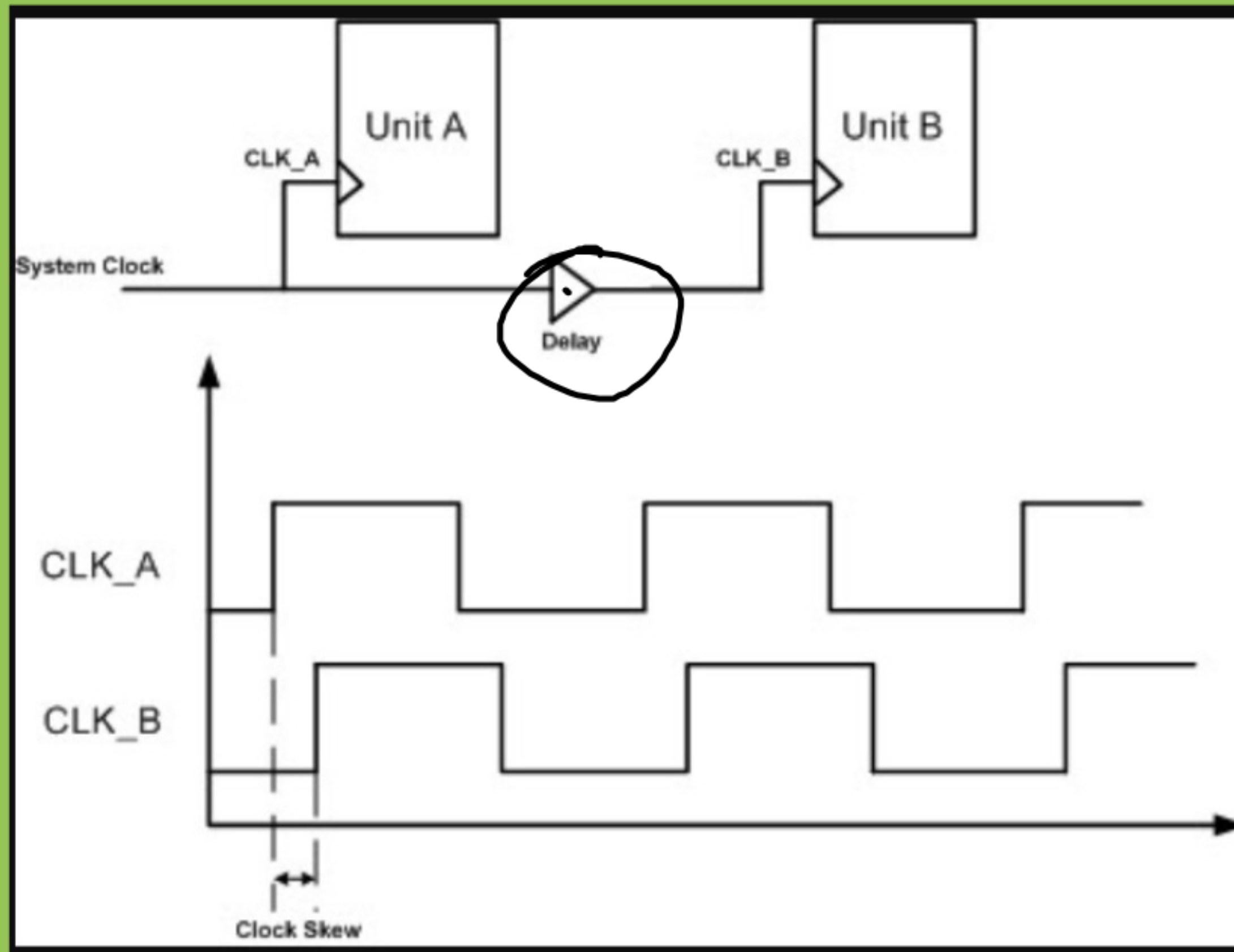


1. Propagation Delay (T_{pd}):

- This is the **dominant delay** and refers to the time it takes for a signal to travel through a single logic gate and reach its final stable state (high or low voltage level) at the output.

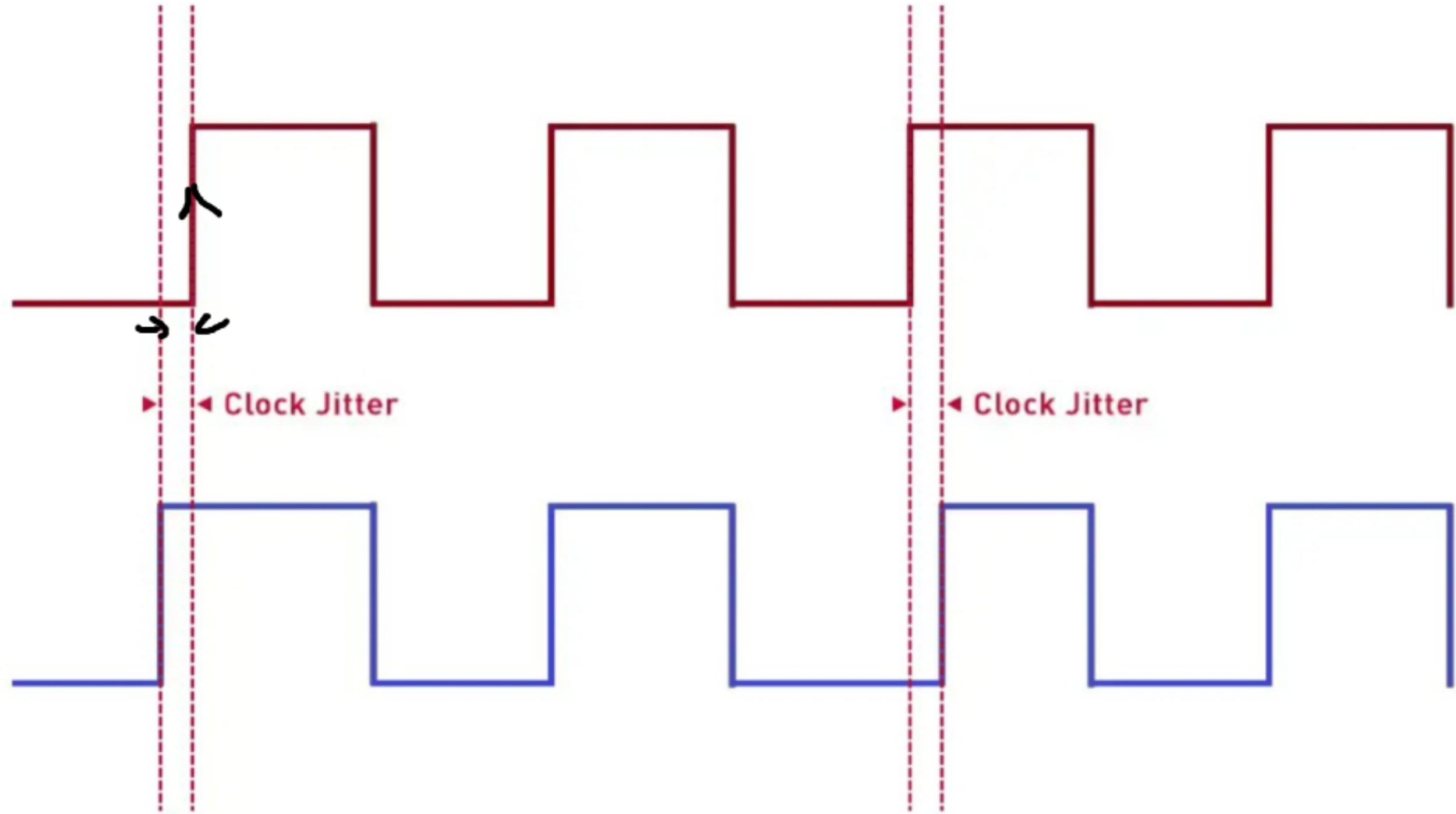
2. Contamination Delay (T_{cd}):

- This is a **shorter delay** compared to the propagation delay.
- It represents the time it takes for the change in the input signal to begin affecting the output voltage level. In simpler terms, it's the time it takes for the output to start transitioning from its current state towards the final state.

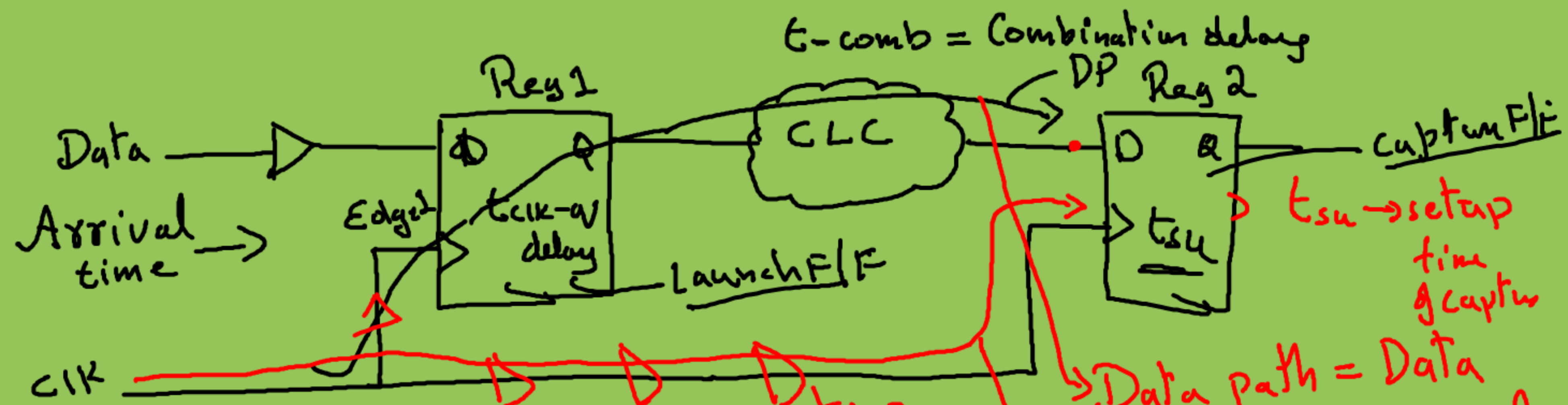


Clock skew
 $= \text{CLK}_B - \text{CLK}_A$

**Ideal
Clock**



**Clock
with
Jitter**



Arrival time: Time taken by the data to reach the input pin of capture flop.

Required time: Time taken by the clock to reach the capture clock.

$$\underline{\text{Slack} = \text{Data arrival time} - \text{Data required time.}}$$

1. Setup slack :

$$\text{Setup time constraints} = \overset{\text{Data Path}}{t_{\text{clk-q}} + t_{\text{pd}} + t_{\text{su}}} \leq \overset{\text{Clock Path}}{t_{\text{clk}} + t_{\text{b}_n}} \quad (\text{Capture flop})$$

$$\text{Setup-slack} = t_{\text{clk}} - t_{\text{clk-q}} + t_{\text{pd}} + t_{\text{su}}(\text{CF})$$

$$\text{Setup-slack} = \underset{\text{CP} - \text{DP}}{\text{DRT} - \text{DAT}} \quad \begin{array}{ll} +ve & \text{no violation} \\ -ve & \text{violation} \end{array}$$

2. Setup slack

2. Hold slack

$$\text{Hold time constraint} = \overbrace{t_{clk-q} + t_{pd}}^{\text{Data path}} \geq \overbrace{t_{hold}}^{\text{clock path}} \left(\begin{array}{l} \text{capture} \\ f/f' \end{array} \right)$$

$$\text{Hold slack} = \text{DAI} - \text{DRT}$$

$$\text{Hold slack} = t_{clk-q} + t_{pd} - t_{add}$$

+ve

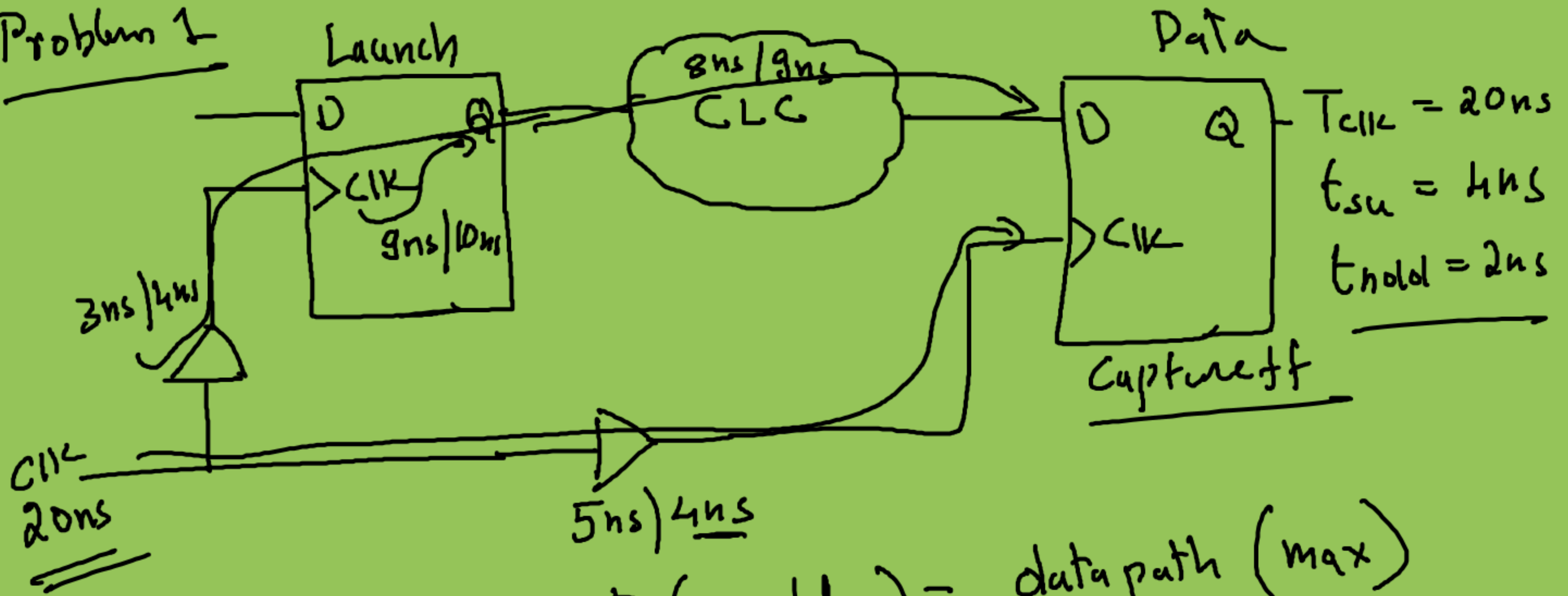
no hold violation

-ve

hold violation occurs

.

Problem 1



Setup time constraint (max delay) = data path (max)
 clock path (min)

Hold time constraint (min delay) = data path (min)
 clock path (max)

Setup time - Constraint $4ns + 10ns + 9ns + 4ns \leq 20ns + 4ns$

$$27ns \leq 24ns.$$

$$\begin{aligned} \text{Setup_slack} &= 24ns - 27ns \\ &= -3ns \end{aligned}$$

setup violation occurred.

Hold time : $3ns + 9ns + 9ns \geq 5ns + 2ns$

$$20 \geq 7ns$$

$$\begin{aligned} \text{Hold slack} &= 20ns - 7ns \\ &= +13ns \end{aligned}$$

Hold violation not occurred.